Sartaaj Khan

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Education

University of Toronto **Sept. 2023 – Present**

PhD Chemical Engineering and Applied Chemistry, Affiliated with Vector Institute
Principal Investigator: Dr. Seyed Mohamad Moosavi; cGPA: 4.00/4.00

University of Toronto **Sept. 2018 – May 2023**

BASc Chemical Engineering and Applied Chemistry, Minor in Bioengineering;
cGPA: 3.73/4.00

Research Experience

Doctoral Researcher **Sept. 2023 – Present**
University of Toronto and Vector Institute for AI *Toronto, ON*

Thesis: Learning the geometry of metal-organic frameworks (MOFs) using data-driven approaches

- Develop multimodal and self-supervised learning models that connect PXRD patterns and precursor text to MOF applications, with fine-tuning across band gap, gas uptake, pore volume, and Henry's coefficient prediction tasks.
- Developed topological graph descriptors of nanoporous bond topology, and benchmark performance against leading graph neural network architectures.
- Contribute to collaborative projects in physics-informed property prediction and MOF-ChemUnity, a curated database linking experimental and computational MOF properties for water-stability and application discovery.
- Mentor undergraduate and graduate researchers in materials informatics, representation learning, and machine-learning workflows.

Undergraduate Researcher **July 2021 – Sept. 2022**
Advanced Membranes Laboratory, University of Toronto *Toronto, ON*

- Built elemental transport models for reverse-osmosis workflows, including ion concentration profiles, water and solute fluxes, recovery ratio, and staged membrane modules.
- Developed and published pySolution, a Python package for non-ideal solution chemistry calculations using the Pitzer model.

Undergraduate Researcher **June 2021 – Sept. 2021**
Ping Lab, University of California, Los Angeles *Los Angeles, CA*

- Developed a bioinformatics pipeline to curate reliable *Sus scrofa* protein sequence databases for cardiac proteomics using UniProt-derived features and cross-organism classification.
- Constructed cardiovascular knowledge graphs and link-prediction workflows to identify candidate protein relationships in cardiac conduction and muscle-contraction processes.

Undergraduate Researcher **May 2020 – Sept. 2020**
Laboratory of Metabolic Systems Engineering, University of Toronto *Toronto, ON*

- Automated genome-scale *E. coli* strain-design simulations to optimize flux toward approximately 70 industrial chemicals using constrained optimization and flux balance analysis.

- Extended mcPECASO simulations of two-stage bioprocesses to assess phenotypic switches in iJO1366 and iML1515 *E. coli* models.

Publications

Selected publications. For full list, please refer to my Google Scholar.

- **Khan, S. T.**, Moosavi, S. M. (2025). Connecting metal-organic framework synthesis to applications using multimodal machine learning. *Nature Communications*, 16(1), 5642.
- Ai, Q.*, **Khan, S. T.***, Barthel, S., Moosavi, S. M. (2025). "Capturing Global Features of Crystals from Their Bond Networks." *AI for Accelerated Materials Design-ICLR 2025*.
- Song, Z., ..., **Khan, S. T.**, ..., & Duan, C. (2025). "Evaluating large language models in scientific discovery." *arXiv* preprint arXiv:2512.15567.
- Jiang, H., **Khan, S. T.**, Werber, J. R., & Karanikola, V. (2025). Modeling scaling prevention and attainable recovery using hypothetical calcium-permeable reverse osmosis membranes. *ACS ES&T Engineering*, 6(1), 303–312.
- Kochi, M. R., Rezaei, H., **Khan, S. T.**, Mamillapalli, B., Ebrahimiazar, M., Ye, H., ... & Moosavi, S. M. (2025). Physics-informed machine learning for predicting temperature-dependent chemical properties. *Digital Discovery*.
- Pruyn, T. M., Aswad, A., **Khan, S. T.**, Huang, J., Black, R., & Moosavi, S. M. (2025). MOF-ChemUnity: Literature-Informed Large Language Models for Metal–Organic Framework Research. *Journal of the American Chemical Society*, 147(47), 43474–43486.
- Zimmermann, Y., Bazgir, A., . . . , **Khan, S. T.**, . . . , & Blaiszik, B. (2025). 32 examples of LLM applications in materials science and chemistry: Towards automation, assistants, agents, and accelerated scientific discovery. *Machine Learning: Science and Technology*, 6(3), 030701.
- Zimmermann, Y., . . . , **Khan, S. T.**, . . . , & Blaiszik, B. (2024). "Reflections from the 2024 large language model (LLM) hackathon for applications in materials science and chemistry." *arXiv* preprint arXiv:2411.15221.

Invited Presentations

- **Khan, S. T.**, Moosavi, S. M. "Accelerated Discovery of Metal–Organic Frameworks through Synthesis-Informed Structure Generation." [Poster]. Accelerate Conference 2025, Toronto, Canada.
- Ai, Q., **Khan, S. T.**, Barthel, S., Moosavi, S. M. "Capturing Global Features of Crystals from Their Bond Networks." [Paper]. AI for Accelerated Materials Design – ICLR, 2025.
- **Khan, S. T.** "Unveiling insights of metal-organic frameworks using a self-supervised multimodal model with only PXRDs and precursors." [Poster]. American Chemical Society (ACS), 2025, San Diego, California.
- **Khan, S. T.**, Moosavi, S. M. "XRayPro: A Self-Supervised Multimodal Model for MOF Application Recommendations Using PXRD and Precursors." [Paper]. AI for Accelerated Materials Design – NeurIPS, 2024, Vancouver, Canada.
- **Khan, S. T.** (2024, October 8). Unveiling Property Insights and Recommending MOF Applications from X-Ray Diffraction Patterns using Self-Supervised Transformer Models. [Presentation]. Canadian Society for Chemical Engineering (CSChE) 2024, Toronto, Canada.
- **Khan, S. T.**, Moosavi, S. M. (2024, August 7). "Using a self-supervised model for property predictions and MOF application recommendations with only powder x-ray diffraction patterns." [Poster]. Accelerate Conference 2024, Vancouver, Canada.

Work Experience

Artificial Intelligence Engineer — **Sept. 2021 – Sept. 2022**

Whitehorse, YK

MGRID Energy Inc.

- Developed YOLO-based vehicle-classification pipelines and designed a hybrid nonlinear-regression/CNN architecture for wildfire detection.
- Curated wildfire-model features from the literature and generated synthetic text data for an NLP bidding-alert prototype.

Teaching Experience

Machine Learning Tutor **Feb. 2025 – Present**
Toronto, ON
University of Toronto

- Taught applied machine-learning concepts for DSI CrossTALK, including clustering, molecular fingerprints, and model selection for drug-discovery workflows.
- Troubleshot Python, data-loading, and memory issues across XGBoost, LightGBM, and related machine-learning tooling during hands-on sessions.

Teaching Assistant **Sept. 2025 – Present**
Toronto, ON
University of Toronto

Courses taught: CHE260: Thermodynamics and Heat Transfer; CHE333: Chemical Reaction Engineering.

Research Mentorship

- Zakariyya Brewster (Summer 2025): University of Toronto undergraduate student in Engineering Science; Common pitfalls in applying machine learning in metal-organic frameworks.
- Hari Om Chadha (Summer 2024): University of Toronto undergraduate student in Engineering Science; Pretraining graph convolutional networks using Barlow-Twin for predicting fluid properties.

Awards

- **Doctoral Student Fellowship – Data Science Institute**, University of Toronto (May 2026). $75,000 over 3 years.
- **Canada Graduate Research Scholarship – Doctoral (CGRS-D)**, NSERC (May 2026 – May 2029). $120,000 over 3 years.
- **ChatGPT 26**, OpenAI (May 2026). One of 26 North American teams selected; $10,000 grant and research support for MOF-ChemUnity and nanoporous-materials research.
- **Microscopy Society of America Award** (Feb. 2026). Best all-student team globally, Microscopy Hackathon 2026; $500.
- **Canada Regional Award**, Elsevier Computational Materials Science (Feb. 2026). Ranked 1st in Canada, Microscopy Hackathon 2026; $500.
- **Queen Elizabeth II/DuPont Canada Scholarship in Science and Technology** (Sept. 2025 – Sept. 2026). $15,000.
- **University of Toronto Fellowship** (Sept. 2023 – Sept. 2025). $13,200.
- **LLMs in Chemistry and Materials Hackathon** (May 2024). 3rd place; $250 prize funded by Radical AI and Anthropic.
- **Bayesian Optimization for Chemistry and Materials Hackathon** (March 2024). 2nd place; $500 CAD prize funded by Acceleration Consortium.
- **Dean's List**, University of Toronto (Sept. 2018 – May 2023). Awarded for all undergraduate semesters.
- **NSERC USRA** (May 2020 – Sept. 2020). $6,000 CAD undergraduate research award.

Skills

Programming:	Python, MATLAB and Simulink, Cypher Query Language, Neo4j, HTML, CSS, Bash
Machine learning:	PyTorch, TensorFlow, Scikit-learn, PyTorch Geometric (PyG), Deep Graph Library (DGL), Torchvision
Software:	ChemDraw, AutoCAD, Aspen Plus, Aspen Dynamics, VESTA, iRASPA
Applications:	Microsoft Word, Microsoft PowerPoint, Microsoft Excel, Microsoft Visio